



HR Overtime Management System

Technical & Functional Documentation — Odoo 19

Modules: hr_overtime_management + hr_overtime_payroll (optional)

Version: 19.0.1.0.3

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Database: odoo19



1. Executive Summary

This document describes the custom Odoo 19 overtime management solution delivered for your organization. The system allows employees to submit overtime requests linked to projects and tasks, routes them through a multi-level approval chain (department manager → optional upper manager → HR), calculates overtime cost from the employee contract wage, optionally creates timesheet lines on final approval, and optionally pushes costs to payroll.

The implementation follows Odoo best practices: reusable approval engine (mirroring hr_holidays patterns), configurable pay-rate multipliers, no duplicate time-tracking tables, and optional payroll integration in a separate glue module.

2. Business Need & Objectives

The following business requirements drove this development:

- Employees submit overtime with project/task allocation and supporting documents.
- Approval follows the existing organizational hierarchy (manager chain + mandatory HR).
- Overtime pay rates (Regular 1.5×, Weekend 2.0×, Holiday 2.5×) are configurable without code changes.
- Approved overtime can generate account.analytic.line (timesheet) entries automatically.
- Overtime cost is derived from contract wage, not a parallel payroll engine.
- A reusable approval-chain component supports future request types (expenses, permissions, etc.).
- Security: employees see own requests; managers see pending approvals; HR sees all.

3. Delivered Modules

Module	Type	Purpose	Dependencies
hr_overtime_management	Core (required)	Requests, approval workflow, reporting, settings	hr, hr_timesheet, project, mail, resource
hr_overtime_payroll	Optional glue	Push approved overtime to payslip inputs	hr_overtime_management, hr_payroll

4. System Architecture

4.1 Reusable Approval Engine

Component	Model	Role
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Chain Service	hr.approval.chain.service	Resolves manager → upper manager → HR chain from employee.parent_id
Chain Mixin	hr.approval.chain.mixin	Generic submit / approve / refuse workflow with mail activities
Overtime Request	hr.overtime.request	Concrete implementation using the mixin

4.2 Approval Chain Logic

Mirrors Odoo hr_holidays manager resolution pattern:

- Step 1: employee.parent_id (or department manager if no parent) → dept_manager
- Step 2: parent of dept manager (if different person) → upper_manager
- Step 3: HR officer from group_overtime_hr_officer → hr (always mandatory final stage)

Two scenarios:

- 2 stages: Solo manager → dept_manager → HR
- 3 stages: dept_manager → upper_manager → HR

4.3 Workflow States

State	Meaning
draft	Employee is editing; not yet submitted
submitted	Waiting on first approver (dept manager)
manager_approved	Dept manager approved; waiting upper manager or HR
upper_manager_approved	Upper manager approved; waiting HR
hr_approved	Fully approved — triggers timesheet / payroll hooks
refused	Rejected at any stage (reason required)
cancel	Cancelled by employee before completion

5. Database Models & Columns

5.1 hr.overtime.type (Configuration)

Column	Type	Description
name	Char	Display name: Regular Overtime, Weekend, Public Holiday
code	Char	Unique code per company: regular, weekend, holiday
rate_multiplier	Float	Pay multiplier — seeded 1.5 / 2.0 / 2.5, editable in Settings
sequence	Integer	Display order
company_id	Many2one	Company scope
active	Boolean	Archive inactive types



5.2 hr.overtime.request (Main Record)

Column	Type	Description
name	Char	Auto sequence OT/YYYY/MM/00001
employee_id	Many2one	Employee who worked overtime
department_id	Many2one	Related from employee, stored
manager_id	Many2one	Direct manager (parent_id), stored
start_datetime	Datetime	Overtime start (date & time picker)
end_datetime	Datetime	Overtime end (date & time picker)
date	Date	Computed from start_datetime for filters/reports
overtime_hours	Float	Computed: (end – start) in hours
overtime_type_id	Many2one	Rate category with multiplier
project_id	Many2one	Project (timesheet-enabled)
task_id	Many2one	Task within project
description	Text	Reason / details
attachment_ids	Many2many	Supporting documents
state	Selection	Workflow status (see section 4.3)
approval_line_ids	One2many	Full approval audit trail
current_approver_id	Many2one	User who must act now (record rules)
hourly_cost	Monetary	From contract wage via hr.version
total_cost	Monetary	hours × hourly_cost × rate_multiplier
analytic_line_id	Many2one	Timesheet line created on HR approval
company_id	Many2one	Company
currency_id	Many2one	Company currency

5.3 hr.overtime.approval.line (Audit Trail)

Column	Type	Description
request_id	Many2one	Parent overtime request
sequence	Integer	Order in chain (10, 20, 30...)
role	Selection	dept_manager upper_manager hr
approver_id	Many2one	Assigned approver user
state	Selection	pending to_approve approved refused
decision_date	Datetime	When decision was made
comment	Text	Refusal reason or notes

5.4 res.company Extensions

Column	Type	Default	Description
overtime_generate_analytic_line	Boolean	True	Auto-create timesheet on HR approval
overtime_default_type_id	Many2one	Regular	Default overtime type for new requests



overtime_daily_hours_cap	Float	4.0	Warning if single request exceeds hours
overtime_hours_per_month	Float	173.33	Fallback divisor: wage ÷ hours for hourly cost

6. Cost Calculation

Formula: $\text{total_cost} = \text{overtime_hours} \times \text{hourly_cost} \times \text{overtime_type.rate_multiplier}$

6.1 Hourly Cost (Odoo 19)

In Odoo 19, contract data lives on `hr.version` (linked via `employee.current_version_id`). Hourly cost is calculated as:

- Primary: $\text{wage} \times 12 \div 52 \div \text{hours_per_week}$ (from employee work schedule)
- Fallback: $\text{wage} \div \text{overtime_hours_per_month}$ (company setting, default 173.33)

Example — Administrator, wage \$7,540/month, 38 h/week, 2 h Regular OT (1.5×): Hourly $\approx \$45.81 \rightarrow \text{Total} \approx \137.43

7. Security Groups & Access

Group	Users	Access
group_overtime_user	All employees	Create/read own requests; edit only in draft
Managers (record rule)	<code>current_approver_id = me</code>	Approve/refuse pending requests
group_overtime_hr_officer	HR team	Read all; final HR approval
group_overtime_admin	Administrators	Configure types, settings, full access

8. User Interface

Menu	Purpose
Overtime → My Requests	Employee self-service list/form
Overtime → My Approvals	Kanban for pending approvers
Overtime → All Requests	HR officers — full list
Overtime → Configuration → Overtime Types	Admin — rate multipliers
Settings → Employees → Overtime	Company configuration

Views delivered: List, Form (statusbar + approval tab + chatter), Kanban, Search filters, Pivot, Graph, QWeb PDF report.

9. Integrations



9.1 Timesheets (account.analytic.line)

When overtime_generate_analytic_line is enabled (default), final HR approval creates a timesheet line with project, task, employee, date, and overtime_hours as unit_amount.

9.2 Payroll (optional — hr_overtime_payroll)

If hr_payroll is installed and overtime_link_to_payroll is enabled, HR approval creates/updates an hr.payslip.input line (type Overtime) on the employee's open payslip.

10. Issues Encountered & Resolutions

Issue	Cause	Resolution
Submit crash: res.groups has no attribute users	Odoo 19 uses all_user_ids not users	Fixed get_hr_responsible() to use all_user_ids
column start_datetime does not exist	Code updated but module not upgraded on odoo19	SQL migration + module upgrade
OwlError: start_time field undefined	Stale ir.ui.view records in database	Updated stored view XML in odoo19
Approval chain incomplete: no HR stage	Sanitizer removed HR when same user as manager	HR stage always preserved in chain
Hourly/Total cost \$0.00	Core returned 0; Odoo 19 uses hr.version not hr.contract	Read wage from employee.version_id
column overtime_hours_per_month missing	New field without DB upgrade	Added column + reset module state
PDF: wkhtmltopdf not found	Binary not installed on Windows	Installed wkhtmltopdf 0.12.6 via winget

11. Configuration Checklist

- 1. Install hr_overtime_management on database odoo19.
- 2. Assign users to security groups (especially Officer: Overtime HR Approval).
- 3. Verify overtime types and multipliers under Overtime → Configuration.
- 4. Set company settings: default type, daily cap, hours/month divisor, timesheet toggle.
- 5. Ensure employees have contract wage and work schedule on hr.version.
- 6. Optional: install hr_overtime_payroll for payslip integration.
- 7. Optional: add C:\Program Files\wkhtmltopdf\bin to PATH for PDF reports.

12. End-User Guide

12.1 Employee — Submit Overtime



- Go to Overtime → My Requests → New.
- Select employee, start/end date & time, overtime type, project, task, description.
- Attach documents if needed → Save → Submit.
- Track status in form statusbar and Approval History tab.

12.2 Manager — Approve

- Go to Overtime → My Approvals.
- Open pending request → Approve or Refuse (reason required).
- If you are also HR, you may need to approve twice (manager step then HR step).

12.3 HR — Final Approval

- Any user in Officer: Overtime HR Approval can perform final approval.
- On approval: state → Approved; timesheet line created if enabled; payroll input if enabled.

13. Technical File Structure

extra_addons/hr_overtime_management/

- models/hr_approval_chain_service.py — reusable chain resolver
- models/hr_approval_mixin.py — generic workflow mixin
- models/hr_overtime_request.py — main request + state machine
- models/hr_overtime_approval_line.py — approval trail
- models/hr_overtime_type.py — configurable rate types
- security/security.xml — groups + record rules
- views/ — list, form, kanban, search, pivot, graph
- report/hr_overtime_report.xml — QWeb PDF
- wizard/hr_overtime_refuse_wizard.py — refusal reason dialog
- tests/test_hr_overtime.py — automated tests
- migrations/ — database upgrade scripts

14. Upgrade & Maintenance

After any code change, upgrade the module:

```
python odoo-bin -c debian/odoo.conf -d odoo19 -u hr_overtime_management --stop-after-init
```



Or: Apps → HR Overtime Management → Upgrade

Always use database odoo19 (not odoo) for this installation.

— End of Document —